

Counting the symmetric relations.
 → Diagonal elements $\left[\begin{array}{l} \text{include (0)} \\ \text{exclude (1)} \end{array} \right] 2^n$

→ Off-diagonal $\longleftrightarrow$ mandated to close in
 pairs $(a,b) \Rightarrow (b,a)$

No. of pairs $\frac{n^2 - n}{2}$

include a pair (1)
 exclude a pair (0)

$2 \times 2 \dots \times 2$

$= 2^{\frac{n^2 - n}{2}}$

Total choices = $2^n \times 2^{\frac{n^2 - n}{2}} = 2^{\frac{n(n+1)}{2}}$

① ρ : reflexive $a \rho a \quad \forall a \in A$.

② ρ : symmetric $a \rho b \Rightarrow b \rho a \quad \forall a, b \in A$.

③ ρ : transitive.
 $a \rho b, b \rho c \Rightarrow a \rho c \quad \underline{\forall a, b, c \in A}$.

Eg. 1. ~~check if a/b where $\rho = |$~~
~~is transitive?~~

$a \rho b$ means a/b .

Check if ρ is transitive.

$$a/b, b/c \Rightarrow a/c.$$

ρ : antisymmetric ($\neq$ asymmetric).

$$\forall, a, b \in A$$

$$\text{if } (a \rho b, b \rho a) \Rightarrow a = b.$$

Eg. 2. ~~Let~~ Let $a \rho b$ mean $a \leq b$,
 $a, b \in \mathbb{N}$.

check if ρ is antisymmetric.

$$(a \leq b, b \leq a) \Rightarrow a = b.$$

$\subseteq \rightarrow$ antisymmetric.

A relation ρ is called a partial order or a POSET if

(a) ρ is reflexive

(b) ρ is antisymmetric

(c) ρ is transitive

check if $a \leq b$ defines a ~~poset~~ in N .

$a \leq a \Rightarrow$ reflexive.

$a \leq b, b \leq a \Rightarrow a = b \Rightarrow$ antisymmetric

$a \leq b, b \leq c \Rightarrow a \leq c \Rightarrow$ transitive.

All relations that are POSET using the special notation

$\rho \Rightarrow \leq$

⇒ Algorithmic complexity analysis.

$$O \longrightarrow f(n) \leq c g(n)$$

Also is in $O(n) \rightarrow O(n^2) \rightarrow O(n^3) \dots$

$$O(n) \leq O(n^2) \leq O(n^3) \dots$$

O : defines a POSET.

A relation ρ is said to be an equivalence relation if

- (a) ρ is reflexive
- (b) ρ is symmetric
- (c) ρ is transitive.

The relation $a = b$ for $\forall a, b \in \mathbb{N}$ defines an equivalence relation? ✓✓ It does!

$\rho : \cong$ (denote equivalence relation).

Defⁿ

If A, B, C are sets s.t.

$$\rho_1 \subseteq A \times B, \rho_2 \subseteq B \times C$$

Composite relation: $\rho_1 \circ \rho_2$

$\rho_1 \circ \rho_2 = \{ (x, z) \mid x \in A, z \in C \text{ and there exists a bridge element } y \in B \text{ such that } (x, y) \in \rho_1 \text{ \& } (y, z) \in \rho_2 \}$.

Eg: $\rho_1 = \{ (1, x), (2, x), (3, y), (3, z) \}$

$$\rho_2 = \{ (w, 5), (x, 6) \}$$

$$\rho_1 \circ \rho_2 = \{ (1, 6), (2, 6) \}$$

Theorem:

Let A, B, C, D be sets s.t.
 $\rho_1 \subseteq A \times B$, $\rho_2 \subseteq B \times C$ & $\rho_3 \subseteq C \times D$.

$$\rho_1 \circ (\rho_2 \circ \rho_3) = (\rho_1 \circ \rho_2) \circ \rho_3.$$

Let $(a, d) \in \rho_1 \circ (\rho_2 \circ \rho_3)$.

For some $b \in B$

$(a, b) \in \rho_1$ and $(b, d) \in \rho_2 \circ \rho_3$.

Again for some $c \in C$

$(b, c) \in \rho_2$ & $(c, d) \in \rho_3$.

$(a, b) \in \rho_1$ & $(b, c) \in \rho_2$

$\Rightarrow (a, c) \in \rho_1 \circ \rho_2$

$(a, c) \in \rho_1 \circ \rho_2$ and $(c, d) \in \rho_3$

1. $(a, d) \in (\rho_1 \circ \rho_2) \circ \rho_3$.

$$\frac{\rho_1 \circ \rho_2}{\quad} \circ \rho_3$$

↓
Makes the
composition
parenthesis
free.

Defⁿ

0-1 matrix.

An $m \times n$ zero-one matrix $E = (e_{ij})_{m \times n}$ is a rectangular array of numbers arranged in m rows & n columns where each

e_{ij} for $1 \leq i \leq m$ & $1 \leq j \leq n$ denotes the i^{th} row & the j^{th} column of E s.t. each entry is either 0 or 1

E.S.

$$E = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix}_{3 \times 4}$$

$$e_{11} = 1, e_{23} = 0$$

→ follows the same rules as
of ~~matrix~~ matrix addition &
multiplication with the
only exception that ~~add~~

$$\boxed{1 + 1 = 1.}$$

Defⁿ Given a set A & a relation ρ
 we define powers of ρ recursively by
 (a) $\rho^1 = \rho$ (b) $\rho^{n+1} = \rho \circ \rho^n$
 $n \in \mathbb{Z}^+$

E.g. $A = \{1, 2, 3, 4\}$.

$$\rho = \{ (1, 2), (1, 3), (2, 4), (3, 2), (3, 4) \}.$$

$$\rho \circ \rho = \rho^2 = \{ (1, 4), (1, 2), (3, 4) \}.$$

$$\rho^3 = \{ (1, 4) \}.$$

$$\rho^4 = \underline{\emptyset}$$

$$\rho^n \quad (n \geq 4) = \underline{\emptyset}$$

Representing relations as 0-1 matrices.

$$P_1 = \{(\underline{1}, x), (2, x), (3, y), (3, z)\}$$

$$P_2 = \{(w, 5), (x, 6)\}$$

$$M(P_1) = \begin{matrix} & \begin{matrix} w & x & y & z \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \end{matrix}$$

$$M(P_2) = \begin{matrix} & \begin{matrix} 5 & 6 & 7 \end{matrix} \\ \begin{matrix} w \\ x \\ y \\ z \end{matrix} & \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \end{matrix}$$

$$P_1 \circ P_2 = ?$$

$$M(P_1) \times M(P_2) = M(P_1 \circ P_2)$$

↓
matrix multiplication
(1+1=1)

$$A = \{1, 2, 3, 4\}, \rho = \{(1, 2), (1, 3), (2, 4), (3, 2)\}$$

$$M(\rho) = \begin{array}{c|cccc} & 1 & 2 & 3 & 4 \\ \hline 1 & 0 & 1 & 1 & 0 \\ 2 & 0 & 0 & 0 & 1 \\ 3 & 0 & 1 & 0 & 0 \\ 4 & 0 & 0 & 0 & 0 \end{array}$$

$$M(\rho^2) = [M(\rho)]^2$$

a power exists as long as the matrix is non-null.

Defⁿ

$$E = (e_{ij})_{m \times n} \text{ \& \& } F = (f_{ij})_{m \times n}$$

E precedes F or E is less than F or $E \leq F$

if $e_{ij} \leq f_{ij}$ for all $1 \leq i \leq m$
 $1 \leq j \leq n$

Eg: $E = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix} \times F = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix} \rightarrow \text{then } E \leq F.$

Defⁿ

For $n \in \mathbb{Z}^+$

$$I_n = (\delta_{ij})_{n \times n}$$

$\rightarrow n \times n$ is $(0,1)$ matrix

s.t.

$$\delta_{ij} = \begin{cases} 1 \\ 0 \end{cases}$$

$\begin{cases} i=j \\ \text{otherwise} \end{cases}$

Identity
matrix

Defⁿ

$A = (a_{ij})_{m \times n}$ be a $(0,1)$ -matrix

The transpose of A , A^T is the matrix

$(a_{ji}^*)_{n \times m}$ where $a_{ji}^* = a_{ij}$

for all $\begin{matrix} 1 \leq i \leq m \\ 1 \leq j \leq n \end{matrix}$

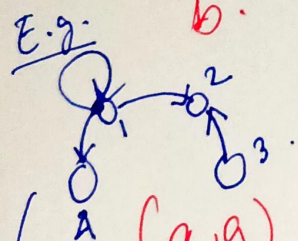
E.g.

$$A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \\ 1 & 1 \end{bmatrix} \Rightarrow A^T = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

Defⁿ

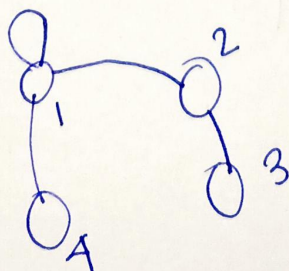
Let V be a finite nonempty set.
A directed graph (digraph) G on V is made
up of the elements of V , called the
vertices/nodes of G & a subset E of $V \times V$
called (directed) edges or arcs of G .

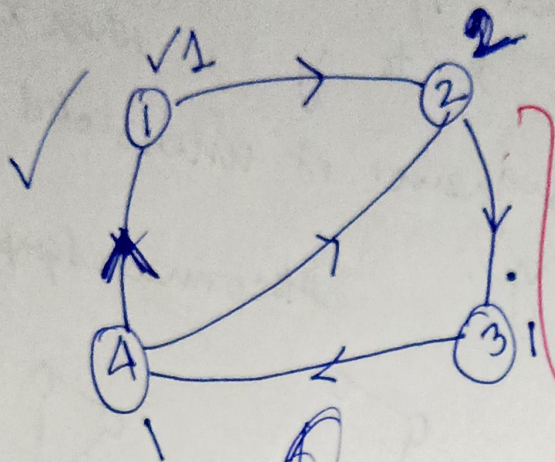
If $a, b \in V$ & $(a, b) \in E$ then we
say that there is an edge from a to
 b .
 $a \rightarrow$ source / origin vertex
 $b \rightarrow$ sink / terminus vertex.



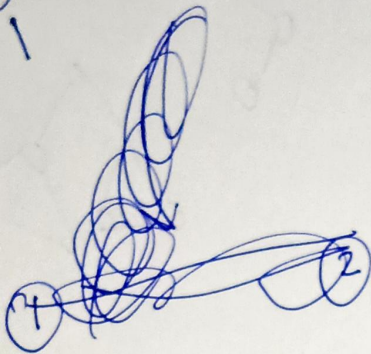
(a, a) stands for a loop in the graph.

If the directions are ignored, we obtain
the corresponding undirected graph.

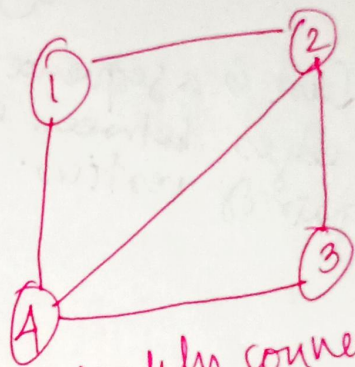




Is this strongly connected?



Remove all arrows / directions.

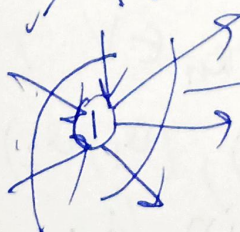
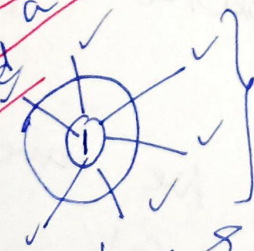


Weakly connected graph.

degree = 6.

degree (no. of edges connecting a node).

A graph with a node having in-degree zero can never have a strongly connected component.

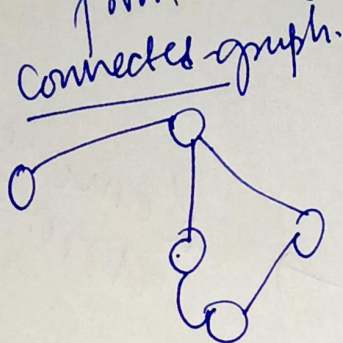


out-degree.

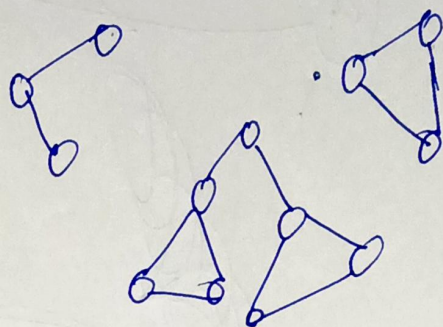
in-degree = 3.
out-degree = 3.

Defⁿ

A graph is called connected if there is a path from every vertex x to y , where $x \neq y$.
Such a path $\rightarrow$ sequence of undirected edges from x to y .



Disconnected graph.



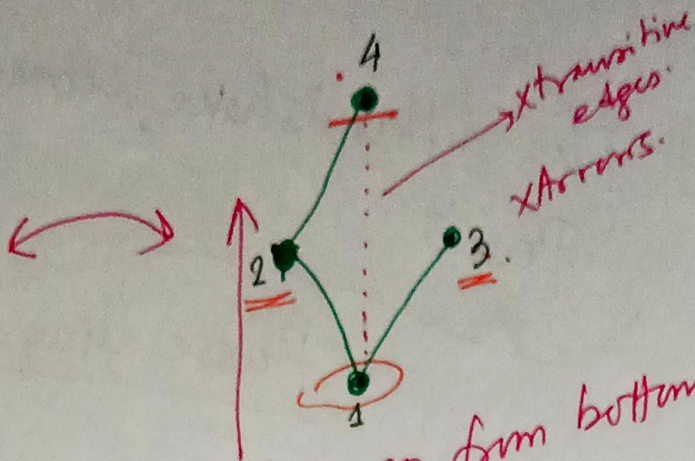
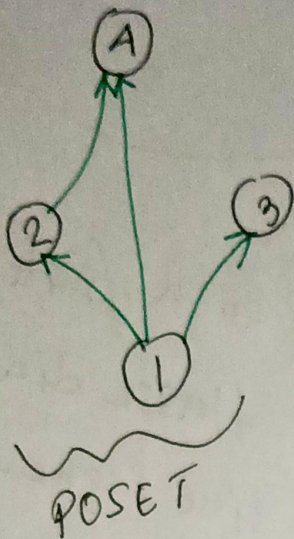
There is a sequence of edges between every pair of vertices.

Defⁿ

A directed graph is called strongly connected if for all vertices $x, y \in V$, where $x \neq y$ there is a path of directed edges from x to y .

① $(x, y) \in E$

② for some $n \in \mathbb{Z}^+$ we have distinct vertices $v_1, v_2, \dots, v_n \in V$, the directed edges ~~$(v_1, v_2), (v_2, v_3), \dots, (v_{n-1}, v_n)$~~ $(x, v_1), (v_1, v_2), \dots, (v_n, y)$ are in G .



directions go from bottom to top
a covers b.

$b \leq a$
covers.
a is above b in the Hasse diagram.

E.g. $\mathcal{U} = \{1, 2, 3\}$.

$\mathcal{P}(\mathcal{U}) = \{ \emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\} \}$.

$\leq \longrightarrow \subseteq$ (subset relationship)

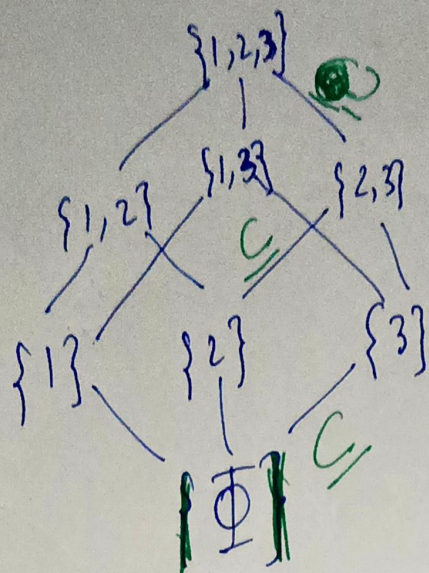
Draws the Hasse diagram.

The Handshake between digraphs & Posets

Hasse diagram.

$\leq$ is a partial order on a finite set A , then we construct a Hasse diagram of $\leq$ by drawing a line segment x up to y if $x, y \in A$ with $x \leq y$ and most importantly there is no other element $z \in A$ s.t. $x \leq z$ and $z \leq y$. The Hasse diagram is always read from bottom to top making the directions implicit.

Ans.



Defⁿ If $(A, \leq)$ is a poset we say that A is a totally ordered set if for all $x, y \in A$ either $x \leq y$ or $y \leq x$.

~~Ex~~ E.g. On the set of natural nos. $\mathbb{N}$, the mathematical ~~symbol~~ symbol $\leq$ defines a total order. ($x \leq y$)

